

Exact Packet-Pair Correction and a Physical Two-Step Eigenvalue Count

Rigorous Transfer from a Stored Feshbach Block to the
2048-Dimensional Perron/Parity-Extracted Matrix

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Abstract

The preceding stored-model analysis proved that a 2048-dimensional packet-complement Feshbach block has no interior complement pole and exactly one eigenvalue in a certified circle. It did not identify that augmented block with the Perron/parity-extracted physical two-step matrix, because the serialized binary64 packet arrays satisfy $WV \approx I_4$ but not the exact algebraic identity $WV = I_4$.

This paper removes that finite-coordinate caveat. Every stored binary64 entry is interpreted as an exact dyadic rational. Exact integer arithmetic forms

$$G = WV, \quad H = G - I_4$$

and proves

$$\|H\|_F \leq 6.50085 \times 10^{-16} < 1.$$

Hence the exact rational correction

$$\widehat{W} = G^{-1}W$$

exists and satisfies $\widehat{W}V = I_4$ identically. With $\widehat{Q} = I - V\widehat{W}$, the corrected packet and complement are exact oblique projectors for the stored physical two-step matrix $A_{\text{phys}} = U^2$.

The correction is transferred through the complete stored factor graph. Rigorous bounds give

$$\|\widehat{W} - W\|_2 \leq 9.27218 \times 10^{-16}, \quad \|\widehat{Q} - Q\|_2 \leq 1.05400 \times 10^{-15},$$

and

$$\|\widehat{B} - B\|_2 \leq 4.70308 \times 10^{-15}.$$

On every one of the 949 rational boundary leaves inherited from the certified complement atlas, the resulting complement Neumann product is below 3.76781×10^{-9} .

For the Feshbach map, the complete RH-28 primal-dual remainder $\eta_\ell + M_\ell c_\ell$ is retained on each leaf. The stored Feshbach inverse is then transported to the corrected map by a second matrix Rouché homotopy. Its largest rigorous product is

$$0.547184078 < 1.$$

Consequently the corrected complement count remains zero and the corrected Feshbach winding remains one.

Exact coordinate algebra finally gives

$$z^4 \det(zI - A_{\text{phys}}) = \det(zI - \widehat{B}) \det \widehat{F}(z).$$

The exact stored circle excludes zero. Therefore

$$\boxed{N_{\Gamma}(A_{\text{phys}}) = 1}.$$

Thus the exact finite Perron/parity-extracted physical two-step matrix has exactly one eigenvalue in the certified circle, counted algebraically.

The theorem concerns one finite matrix defined by stored binary64 factors. It does not enclose discretization error, prove a zero-noise or dimension limit, construct a self-adjoint operator, identify zeta zeros, or imply the Riemann hypothesis.

Keywords: packet biorthogonalization; Feshbach map; physical two-step operator; matrix Rouché theorem; verified numerical linear algebra; spectral count; nonnormal matrix.

MSC 2020: 15A18; 47A10; 47A11; 47A55; 65F35; 65G20; 65P30.

1 Introduction

A packet–complement Feshbach reduction becomes an exact coordinate description of a physical matrix only when its synthesis and analysis pair is exactly biorthogonal. If $V \in \mathbb{C}^{n \times m}$ and $W \in \mathbb{C}^{m \times n}$ satisfy

$$WV = I_m,$$

then $P = VW$ and $Q = I - P$ are complementary oblique projectors. The packet, forcing, observation, and complement blocks of a matrix A are then exact coordinates of A , and the Feshbach determinant counts physical eigenvalues after the complement poles are removed [9, 12].

In floating construction, the canonical dual W is obtained from a small Gram solve, so the computed product WV is indistinguishable from the identity at ordinary precision. Once the arrays are serialized and every binary64 entry is regarded as an exact number, however, the matrix product is no longer exactly the identity. This distinction was deliberately preserved in the outward-rounded chain: $Q = I - VW$ was used only as a stored linear factor, with no silent assumption that $Q^2 = Q$ [8, 11].

That caution produced an exact but intermediate theorem. RH-34 certified, for the stored block realization,

$$N_{\Gamma}(B) = 0, \quad N_{\Gamma}(\mathcal{M}_{\text{st}}) = 1,$$

and showed that the stored Feshbach determinant has one interior zero [10]. Because $WV = I_m$ was not exact, the theorem did not identify $\mathcal{M}_{\text{st}}$ with the original Perron/parity-extracted physical two-step matrix.

The defect is only of roundoff scale, but size alone is not a proof. The complement resolvent is strongly nonnormal and reaches a rigorous upper bound of 8.01138×10^5 on the counting circle. A perturbation of size 10^{-15} may therefore be harmless, but the coordinate correction must be propagated through both the complement inverse and the Feshbach map. The latter also requires the complete primal–dual remainder, not only the directly computable correction term.

This paper makes that propagation explicit. It first corrects the dual packet exactly by

$$\widehat{W} = (WV)^{-1}W.$$

No data are refitted and no packet range is changed. This is a finite-dimensional coordinate correction applied to the exact stored arrays. Structured norm identities then bound the changes in all four Feshbach blocks. The RH-33 complement atlas transfers the pole-free count, while the RH-28 projected inverse and complete remainder transfer the Feshbach winding. Exact coordinate algebra then converts the corrected relative count into an eigenvalue count for the physical two-step matrix.

The contributions are:

1. an exact dyadic computation of the 4×4 defect $WV - I_4$;
2. a rigorous exact biorthogonal correction $\widehat{W} = (WV)^{-1}W$;
3. structured perturbation bounds for the corrected packet, complement, forcing, observation, and direct blocks;
4. a 949-leaf corrected-complement homotopy using the certified RH-33 resolvent atlas;
5. a 949-leaf Feshbach Rouché comparison retaining the full RH-28 remainder $\eta_\ell + M_\ell c_\ell$;
6. an exact determinant identity between the corrected block and $U^2 \oplus 0_4$; and
7. the rigorous physical count $N_\Gamma(U^2) = 1$.

The evidence hierarchy is again explicit.

- The packet correction, projector identities, block factorization, determinant identity, and count deduction are exact algebra.
- Dyadic Gram products are exact rational computations. Block norms and leafwise products are rigorous outward-rounded bounds.
- The displayed physical eigenvalue plot is a floating diagnostic and is not used to prove the count.

2 Exact stored physical matrix and packet defect

2.1 Perron/parity-extracted two-step matrix

Fix $\sigma = 10^{-2}$ and $n = 2048$. Let $M \in \mathbb{R}^{n \times n}$ be the stored sparse folded Gaussian matrix, and let $R_p, L_p \in \mathbb{R}^{n \times 2}$ and $\Lambda \in \mathbb{R}^{2 \times 2}$ be the stored Perron/parity factors and values. Define

$$U = M - R_p \Lambda L_p^\top, \quad A_{\text{phys}} = U^2. \quad (1)$$

Every factor in [equation \(1\)](#) is an exact stored binary64 array. The matrix A_{phys} is the finite Perron/parity-extracted physical two-step matrix considered by the contour construction. The adjective *physical* refers to this full n -dimensional finite matrix, not to a continuum limit.

Let $V \in \mathbb{R}^{n \times m}$, $W \in \mathbb{R}^{m \times n}$, $m = 4$, be the stored packet synthesis and analysis arrays. Put

$$G = WV, \quad H = G - I_m. \quad (2)$$

In exact real arithmetic the canonical construction is designed to satisfy $WV = I_m$. The serialized arrays have a nonzero exact dyadic H .

2.2 Exact dyadic Gram computation

Each binary64 number is a rational integer times a power of two. The implementation converts every component of W and V to that exact fraction, forms all 16 dot products in WV with integer arithmetic, and subtracts the exact identity. No floating matrix product is used in this step.

The resulting Frobenius square

$$\sum_{i,j=1}^4 H_{ij}^2$$

is an exact rational number. A 256-bit Arb square root rounded upward gives

$$\|H\|_2 \leq \|H\|_F \leq h := 6.500848174495279 \times 10^{-16} < 1. \quad (3)$$

The complete numerators and denominators of all 16 entries are committed in the exact packet-defect ledger.

Lemma 2.1 (Exact packet correction). *The exact matrix $G = I_m + H$ is invertible. Define*

$$\widehat{W} = G^{-1}W, \quad \widehat{P} = V\widehat{W}, \quad \widehat{Q} = I - \widehat{P}. \quad (4)$$

Then

$$\widehat{W}V = I_m, \quad \widehat{P}^2 = \widehat{P}, \quad \widehat{Q}^2 = \widehat{Q}, \quad \widehat{P}\widehat{Q} = \widehat{Q}\widehat{P} = 0. \quad (5)$$

Moreover,

$$\|G^{-1}\|_2 \leq \frac{1}{1-h}. \quad (6)$$

Proof. The Neumann lemma and [equation \(3\)](#) prove invertibility and [equation \(6\)](#). Then $\widehat{W}V = G^{-1}WV = G^{-1}G = I_m$. The remaining identities follow by direct multiplication. $\square$

Remark 2.2 (Nature of the correction). The synthesis range $\text{ran } V$, the physical matrix A_{phys} , and every stored input remain unchanged. The correction only replaces the rounded dual coordinates by the exact dual of the already stored packet basis. Its entries are exact rationals and need not be binary64 numbers.

3 Stored and corrected Feshbach systems

3.1 Stored blocks

The stored external factor is

$$Q = I - VW. \quad (7)$$

Without assuming $Q^2 = Q$, define

$$D = WA_{\text{phys}}V, \quad C = QA_{\text{phys}}V, \quad E = WA_{\text{phys}}Q, \quad B = QA_{\text{phys}}Q. \quad (8)$$

For $R(z) = (zI - B)^{-1}$, the stored Feshbach map is

$$F(z) = zI_m - D - ER(z)C. \quad (9)$$

RH-34 proves on the selected circle Γ that

$$N_\Gamma(B) = 0, \quad \text{wind}_\Gamma \det F = 1. \quad (10)$$

3.2 Corrected blocks

Using the exact pair in [equation \(4\)](#), define

$$\widehat{D} = \widehat{W}A_{\text{phys}}V, \quad \widehat{C} = \widehat{Q}A_{\text{phys}}V, \quad \widehat{E} = \widehat{W}A_{\text{phys}}\widehat{Q}, \quad \widehat{B} = \widehat{Q}A_{\text{phys}}\widehat{Q} \quad (11)$$

and

$$\widehat{F}(z) = zI_m - \widehat{D} - \widehat{E}(zI - \widehat{B})^{-1}\widehat{C}. \quad (12)$$

These are now genuine packet/complement coordinates of A_{phys} .

3.3 Exact coordinate determinant

Set

$$\widehat{K} = \begin{pmatrix} \widehat{W} \\ \widehat{Q} \end{pmatrix} \in \mathbb{C}^{(m+n) \times n}, \quad \widehat{J} = \begin{pmatrix} V & \widehat{Q} \end{pmatrix} \in \mathbb{C}^{n \times (m+n)}. \quad (13)$$

By [equation \(5\)](#),

$$\widehat{J}\widehat{K} = V\widehat{W} + \widehat{Q}^2 = \widehat{P} + \widehat{Q} = \mathbf{I}_n. \quad (14)$$

The corrected augmented block is

$$\widehat{\mathcal{M}} = \widehat{K}A_{\text{phys}}\widehat{J} = \begin{pmatrix} \widehat{D} & \widehat{E} \\ \widehat{C} & \widehat{B} \end{pmatrix}. \quad (15)$$

Proposition 3.1 (Physical block plus coordinate zeros). *The corrected augmented block is similar to $A_{\text{phys}} \oplus 0_m$. Consequently*

$$\det(z\mathbf{I}_{n+m} - \widehat{\mathcal{M}}) = z^m \det(z\mathbf{I}_n - A_{\text{phys}}). \quad (16)$$

Wherever $z\mathbf{I} - \widehat{B}$ is invertible,

$$z^m \det(z\mathbf{I}_n - A_{\text{phys}}) = \det(z\mathbf{I}_n - \widehat{B}) \det \widehat{F}(z). \quad (17)$$

Proof. Equation [equation \(14\)](#) gives the direct sum

$$\mathbb{C}^{m+n} = \text{ran } \widehat{K} \oplus \ker \widehat{J}.$$

The corrected block vanishes on $\ker \widehat{J}$. On $\text{ran } \widehat{K}$,

$$\widehat{\mathcal{M}}\widehat{K}x = \widehat{K}A_{\text{phys}}\widehat{J}\widehat{K}x = \widehat{K}A_{\text{phys}}x.$$

Thus its restriction is conjugate to A_{phys} , while the complementary m -dimensional block is zero. This proves [equation \(16\)](#). Taking the Schur complement of $z\mathbf{I}_n - \widehat{B}$ in [equation \(15\)](#) proves [equation \(17\)](#) [4]. $\square$

4 Structured correction majorants

Let

$$\Delta W = \widehat{W} - W, \quad \Delta Q = \widehat{Q} - Q = -V\Delta W.$$

From $G = \mathbf{I} + H$,

$$\Delta W = -G^{-1}HW. \quad (18)$$

Introduce rigorous upper bounds

$$v \geq \|V\|_2, \quad w \geq \|W\|_2, \quad x \geq \|A_{\text{phys}}V\|_2, \quad y \geq \|A_{\text{phys}}Q\|_2, \quad (19)$$

and

$$d \geq \|D\|_2, \quad c \geq \|C\|_2, \quad e \geq \|E\|_2. \quad (20)$$

Put

$$\delta_W = \frac{h}{1-h}w, \quad \delta_Q = v\delta_W. \quad (21)$$

Proposition 4.1 (Structured block correction). *The exact corrected blocks satisfy*

$$\|\hat{D} - D\|_2 \leq \delta_D := \delta_W x, \quad (22)$$

$$\|\hat{C} - C\|_2 \leq \delta_C := v\delta_W x, \quad (23)$$

$$\|\hat{E} - E\|_2 \leq \delta_E := \delta_W y + d\delta_W + \delta_W x\delta_W, \quad (24)$$

$$\|\hat{B} - B\|_2 \leq \delta_B := v\delta_W y + c\delta_W + v\delta_W x\delta_W. \quad (25)$$

Also $\|\hat{C}\|_2 \leq c + \delta_C$.

Proof. The first two bounds follow from

$$\hat{D} - D = \Delta W A_{\text{phys}} V, \quad \hat{C} - C = -V \Delta W A_{\text{phys}} V.$$

Using $\hat{Q} = Q - V \Delta W$,

$$\begin{aligned} \hat{E} - E &= \Delta W A_{\text{phys}} Q - D \Delta W - \Delta W A_{\text{phys}} V \Delta W, \\ \hat{B} - B &= -V \Delta W A_{\text{phys}} Q - C \Delta W + V \Delta W A_{\text{phys}} V \Delta W. \end{aligned}$$

Take induced norms and apply [equation \(21\)](#). □

5 Leafwise transfer of the two count gates

5.1 Corrected complement

Let an inherited RH-33 rational leaf ℓ carry

$$\|(zI - B)^{-1}\|_2 \leq M_\ell \quad (z \in \ell). \quad (26)$$

Define

$$\mu_\ell = M_\ell \delta_B. \quad (27)$$

Lemma 5.1 (Corrected complement resolvent). *If $\mu_\ell < 1$, then $zI - \hat{B}$ is invertible on the leaf and*

$$\|(zI - \hat{B})^{-1}\|_2 \leq \hat{M}_\ell := \frac{M_\ell}{1 - \mu_\ell}. \quad (28)$$

The straight-line homotopy from B to $\hat{B}$ is boundary invertible on that leaf.

Proof. Factor

$$zI - \hat{B} = (zI - B) \left[I - (zI - B)^{-1} (\hat{B} - B) \right]$$

and apply the Neumann lemma. □

5.2 Stored Feshbach inverse

For the RH-28 parent arc containing ℓ , let κ_ℓ bound the projected Feshbach inverse, η_ℓ be the computable correction ratio, and c_ℓ the primal–dual remainder coefficient. The complete inherited bound is

$$r_\ell = \eta_\ell + M_\ell c_\ell. \quad (29)$$

RH-33 proves M_ℓ lies below the RH-28 conditional budget, hence $r_\ell < 1$ on every refined leaf [\[7, 8\]](#). Therefore

$$\|F(z)^{-1}\|_2 \leq K_\ell := \frac{\kappa_\ell}{1 - r_\ell}, \quad z \in \ell. \quad (30)$$

Remark 5.2 (Why the remainder matters). The archived field called the correction ratio is η_ℓ , not the full quantity in [equation \(29\)](#). Omitting $M_\ell c_\ell$ would not change the numerical conclusion at the worst RH-35 leaf, but it would make the inverse ledger logically incomplete. The present certificate includes both terms with outward arithmetic.

5.3 Corrected Feshbach map

The resolvent identity gives

$$\hat{R} - R = \hat{R}(\hat{B} - B)R, \quad \hat{R} = (zI - \hat{B})^{-1}. \quad (31)$$

Writing $\hat{C} = C + \Delta C$, and similarly for the other blocks,

$$\hat{E}\hat{R}\hat{C} - ERC = (\hat{E} - E)\hat{R}\hat{C} + E(\hat{R} - R)\hat{C} + ER(\hat{C} - C). \quad (32)$$

Thus [proposition 4.1](#) and [lemma 5.1](#) imply

$$\begin{aligned} \|\hat{F} - F\|_2 &\leq \phi_\ell := \delta_D + \delta_E \widehat{M}_\ell(c + \delta_C) \\ &\quad + e\widehat{M}_\ell \delta_B M_\ell(c + \delta_C) + eM_\ell \delta_C. \end{aligned} \quad (33)$$

Theorem 5.3 (Leafwise exact-pair transfer). *If every leaf satisfies*

$$\mu_\ell < 1, \quad q_\ell := K_\ell \phi_\ell < 1, \quad (34)$$

then

$$N_\Gamma(\hat{B}) = N_\Gamma(B), \quad \text{wind}_\Gamma \det \hat{F} = \text{wind}_\Gamma \det F. \quad (35)$$

Proof. The first gate and [lemma 5.1](#) keep the complete complement homotopy boundary-invertible. For the Feshbach maps,

$$\hat{F} = F \left[I + F^{-1}(\hat{F} - F) \right],$$

and [equations \(30\)](#) and [\(33\)](#) give $\|F^{-1}(\hat{F} - F)\|_2 \leq q_\ell < 1$. The straight-line matrix homotopy is therefore boundary-invertible on each leaf. The exact rational leaves cover all of Γ , so spectral-count and determinant-winding invariance give [equation \(35\)](#) [1, 2]. $\square$

6 Rigorous computation

6.1 Outward block bounds

The exact dyadic Gram computation supplies h , while exact 4×4 Gram matrices for $V^\top V$ and $WW^\top$ give rigorous small-matrix upper bounds for v and w . The remaining quantities are computed by the RH-27 componentwise factor graph [11].

The graph evaluates $A_{\text{phys}}V$, D , C , and E directly from the stored sparse and low-rank factors. To bound $A_{\text{phys}}Q$, the 2048 columns are divided into eight blocks of width 256. Each block is propagated through

$$X \mapsto U(U(QX))$$

with one outward complex-disc radius per entry. Block Frobenius bounds are combined by an outward Euclidean sum.

[Table 1](#) lists the resulting global bounds.

Table 1: Rigorous exact-pair and structured block-correction majorants.

Quantity	Upper bound
$\ WV - \mathbf{I}_4\ _F$	6.50085×10^{-16}
$\ (WV)^{-1}\ _2$	1.0000000000000001
$v \geq \ V\ _2$	1.136734
$w \geq \ W\ _2$	1.426303
$x \geq \ A_{\text{phys}}V\ _2$	1.536772
$y \geq \ A_{\text{phys}}Q\ _2$	3.520528
$d \geq \ D\ _2$	1.195080
$c \geq \ C\ _2$	1.070344
$e \geq \ E\ _2$	1.651090
δ_W	9.27218×10^{-16}
δ_Q	1.05400×10^{-15}
δ_D	1.42493×10^{-15}
δ_C	1.61976×10^{-15}
δ_E	4.37240×10^{-15}
δ_B	4.70308×10^{-15}

6.2 Leafwise results

All scalar combinations are evaluated with 256-bit Arb arithmetic and rounded outward. The complete transfer ledger has 949 rows. Every corrected-complement product and every corrected-Feshbach product is below one.

Table 2: Worst leafwise transfer quantities. Both maxima occur on parent arc 878, with center identifier `arc_00879`.

Quantity	Rigorous value
Stored complement inverse M_ℓ	801137.544538
Complement product μ_ℓ	3.76781×10^{-9}
Corrected complement inverse $\widehat{M}_\ell$	801137.547556
Stored projected inverse κ_ℓ	102.575354330
Full stored ratio r_ℓ	1.55882×10^{-7}
Stored Feshbach inverse K_ℓ	102.575370320
Corrected Feshbach difference ϕ_ℓ	0.005334459
Feshbach Rouché product q_ℓ	0.547184078
Feshbach denominator $1 - q_\ell$	0.452815922

The largest stored Feshbach inverse over all leaves is approximately 112653.101, but it occurs where the complement inverse is only 222.619. The leafwise ledger is therefore materially sharper than multiplying unrelated global maxima.

6.3 Floating physical spectrum

As a diagnostic, a dense floating eigensolve of A_{phys} finds exactly one eigenvalue inside the circle. Its floating value is

$$-0.0993604146537115 - 0.4442017426458019i,$$

with a floating interior boundary clearance of 0.0144191037. The nearest exterior candidate has clearance 0.0491585001. These numbers are not used in the proof.

RH-35: exact packet-pair correction and physical two-step count

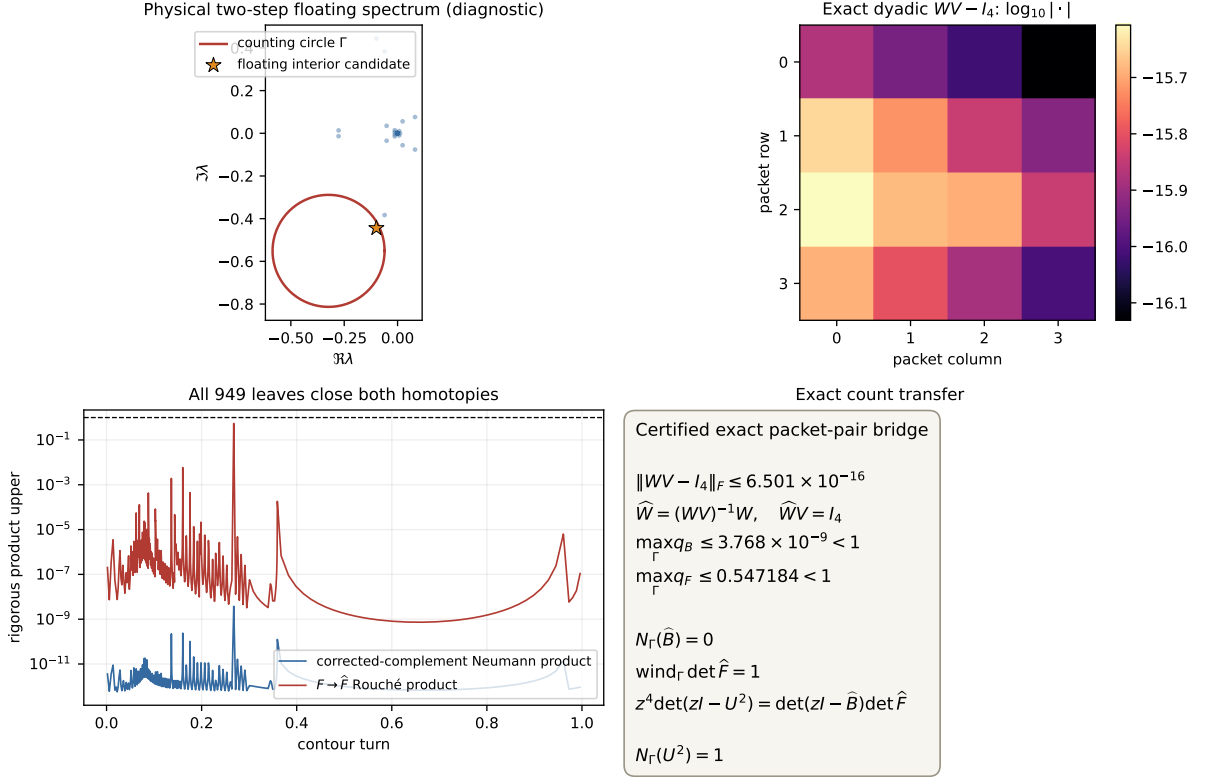


Figure 1: Exact packet-pair transfer. Top left: floating spectrum of the physical two-step matrix, shown only as a diagnostic. Top right: exact dyadic entries of $WV - I_4$. Bottom left: all 949 rigorous corrected-complement and Feshbach products. Bottom right: the exact coordinate and count ledger.

7 Main physical count theorem

Theorem 7.1 (Exact stored physical two-step count). *For the exact 2048×2048 Perron/parity-extracted physical two-step matrix*

$$A_{\text{phys}} = U^2$$

defined by the stored binary64 factors at $\sigma = 10^{-2}$,

$$N_{\Gamma}(A_{\text{phys}}) = 1. \quad (36)$$

Thus exactly one eigenvalue lies in the circle, counted algebraically.

Proof. Table 2 verifies both gates in theorem 5.3 on all 949 leaves. Applying equations (10) and (35) gives

$$N_{\Gamma}(\hat{B}) = 0, \quad \text{wind}_{\Gamma} \det \hat{F} = 1.$$

The exact stored circle has center

$$-0.3233504401504541 - 0.5508412474453575i$$

and radius 0.2624987592858511. Exact dyadic arithmetic gives

$$|c|^2 - r^2 = \frac{27509112334457289433346968324839}{81129638414606681695789005144064} > 0,$$

so zero lies outside the disk and $\text{wind}_\Gamma z^4 = 0$. Taking the winding of [equation \(17\)](#) yields

$$N_\Gamma(A_{\text{phys}}) = N_\Gamma(\widehat{B}) + \text{wind}_\Gamma \det \widehat{F} = 0 + 1.$$

□

Corollary 7.2 (Corrected physical Feshbach zero). *The corrected complement has no eigenvalue in the closed disk, $\widehat{F}$ is holomorphic there, and $\det \widehat{F}$ has exactly one zero in the disk, counted with multiplicity.*

Proof. The complement homotopy excludes boundary crossings and preserves the RH-34 zero interior count. Hence $(zI - \widehat{B})^{-1}$ is holomorphic in the disk. The transferred winding is one, so the ordinary argument principle gives the assertion. □

Remark 7.3 (What the theorem now identifies). The count no longer belongs only to the $(n + m)$ -dimensional stored augmented realization. It belongs to the full n -dimensional matrix $A_{\text{phys}} = U^2$ itself. The theorem still concerns the exact finite serialized discretization; the term *physical* does not mean that a continuum operator limit has been proved.

8 Audit trail and reproducibility

The committed archive contains:

1. the exact rational numerator and denominator of every entry of $WV - I_4$;
2. the exact-pair, block-majorant, leaf-extrema, count, timing, and hash fields in the main certificate JSON;
3. all 949 leaf rows, including the projected inverse, computable RH-28 ratio, remainder coefficient, full inherited ratio, corrected complement product, corrected Feshbach difference, and both homotopy denominators;
4. a floating spectrum pilot clearly labeled nonvalidated;
5. dependency hashes for the RH-28 arc ledger, RH-33 resolvent atlas, RH-34 count theorem, physical model builder, and componentwise arithmetic; and
6. eight unit and archive tests, including a synthetic proof of the $A_{\text{phys}} \oplus 0_m$ coordinate identity.

The rigorous certificate takes approximately 8.0 seconds on the reported server. The exact dyadic Gram stage takes under one second; the dominant step is the blockwise componentwise enclosure of $A_{\text{phys}}Q$. Timings are diagnostics.

The outward arithmetic assumes IEEE binary64 round-to-nearest operations, conservative operation counts, no overflow, and no harmful underflow [3, 6]. Exact Gram products use rational integer arithmetic, and the final positive scalar combinations use 256-bit Arb [5].

9 Scope, limitations, and next gates

The theorem closes the finite-coordinate bridge at one stored scale. It does not establish the following.

1. The stored sparse matrix is not enclosed relative to the exact Gaussian integral kernel or an infinite-dimensional transfer operator.
2. The count has not been continued through smaller noise scales and increasing dimensions.

3. No interval enclosure of the physical eigenvalue location is given. The theorem is a contour count, not a validated eigenvalue box.
4. The floating observation that most eigenvalues have tiny modulus is not an exact rank theorem.
5. No self-adjoint generator, prime-power trace formula, $T \log T$ counting law, zeta-zero identification, Hilbert–Pólya construction, or implication for the Riemann hypothesis is obtained.

The immediate stored-scale chain is now complete:

$$\begin{aligned} & \text{projected winding} \longrightarrow \text{stored Feshbach winding} \\ & \longrightarrow \text{zero complement count} \longrightarrow \text{physical two-step count.} \end{aligned}$$

The next mathematically distinct gates are scale continuation and spectral localization. A natural continuation paper would combine:

1. the exact packet correction at several smaller σ ;
2. a compressed or sparse replacement for dense full-spectrum diagnostics;
3. validated disks for the unique physical eigenvalue;
4. perturbation bounds linking adjacent stored dimensions and noise scales.

Only after such finite-scale continuation would a continuum or zero-noise question become well posed.

10 Conclusion

The stored packet pair missed exact biorthogonality by approximately 6.5×10^{-16} . That discrepancy is tiny, but the preceding papers correctly refused to erase it algebraically. Exact dyadic arithmetic now shows that the packet Gram is invertible and supplies a canonical rational dual correction.

The correction changes the complement block by at most 4.704×10^{-15} . The inherited nonnormal resolvent atlas amplifies this to a worst complement product of 3.768×10^{-9} , still far below one. The more demanding Feshbach comparison retains the complete primal–dual remainder and closes with product 0.547185.

Consequently the corrected packet/complement system has zero complement count and winding one. Exact coordinate factorization then transfers that count to the full stored physical two-step matrix:

$$N_{\Gamma}(U^2) = 1.$$

This is the first point in the chain where the unique contour count belongs to the physical 2048-dimensional finite matrix rather than only to a projected or augmented stored realization. Extending it across scales remains a separate problem.

A Derivation of the block majorants

From $\Delta W = -G^{-1}HW$,

$$\|\Delta W\|_2 \leq \|G^{-1}\|_2 \|H\|_2 \|W\|_2 \leq \delta_W.$$

Because $\Delta Q = -V\Delta W$, $\|\Delta Q\|_2 \leq \delta_Q$. The exact block differences are

$$\begin{aligned}\hat{D} - D &= \Delta W A_{\text{phys}} V, \\ \hat{C} - C &= -V\Delta W A_{\text{phys}} V, \\ \hat{E} - E &= \Delta W A_{\text{phys}} Q - W A_{\text{phys}} V\Delta W - \Delta W A_{\text{phys}} V\Delta W, \\ \hat{B} - B &= -V\Delta W A_{\text{phys}} Q - Q A_{\text{phys}} V\Delta W + V\Delta W A_{\text{phys}} V\Delta W.\end{aligned}$$

These identities explain why the certificate uses the directly enclosed quantities

$$A_{\text{phys}} V, \quad A_{\text{phys}} Q, \quad D, \quad C, \quad E,$$

instead of a crude global bound for $\|A_{\text{phys}}\|$.

B Reproduction

From this paper directory, run

```
PYTHONDONTWRITEBYTECODE=1 python -m pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 python experiments/build_archive.py
MPLBACKEND=Agg python experiments/make_figures.py
python experiments/verify_archive.py
```

The complete rigorous certificate is rebuilt by

```
OPENBLAS_NUM_THREADS=32 OMP_NUM_THREADS=32 \
PYTHONDONTWRITEBYTECODE=1 python \
  experiments/run_packet_pair_certificate.py \
  --sigma 0.01 --chunk-size 256
```

The manuscript is compiled with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

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